

Distributed Systems

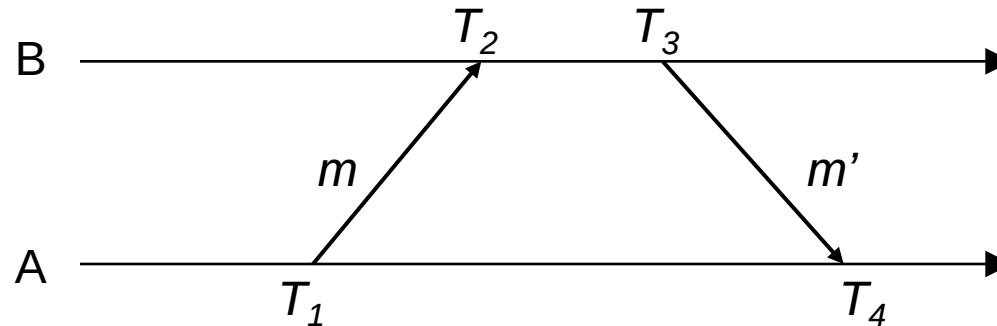
Laboratory:
clock synchronization

Introduction

- We intend to simulate the execution of the NTP symmetric protocol using emails
- For a synchronization experiment two users, A and B, are required, each of them with a clock (of any kind)
- One of the users, A, initiates the protocol and performs the final calculations of deviation and precision
- The rest of the document describes the NTP protocol and the steps to do the experiment. *Cristian's* algorithm is also described, and it is proposed to apply it with the same data, in order to have a basis for comparison

Network Time Protocol (NTP)

Symmetric mode: basics



$$\left. \begin{array}{l} T_2 = T_1 + t + \theta \\ T_4 = T_3 + t' - \theta \end{array} \right\} \begin{array}{l} \text{Hypothesis: A delayed with respect to B } (\theta) \\ t, t' \geq 0 \text{ are the transmission times of messages } m \text{ and } m' \end{array}$$

$$d_i = t + t' = T_2 - T_1 + T_4 - T_3 = (T_4 - T_1) - (T_3 - T_2)$$

$$\theta = \theta_i + (t' - t) / 2, \text{ where } \theta_i = (T_2 - T_1 + T_3 - T_4) / 2$$

$$\theta_i - (d_i / 2) \leq \theta \leq \theta_i + (d_i / 2)$$

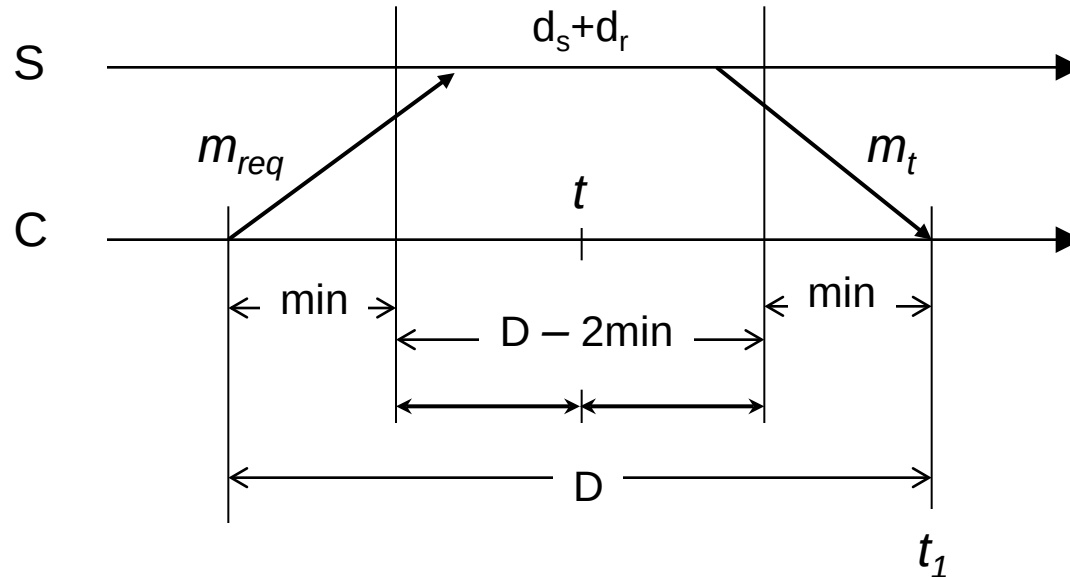
θ_i deviation estimate; $(d_i / 2)$ precision

Network Time Protocol (NTP)

Symmetric mode: experiment

1. A prepares a request message (m) for B. Just before sending the message to B, A checks its local time (T_1) and writes it down
2. B checks the arrival of the request message from A. As soon as he receives it, B checks its local time (T_2) and writes it down
3. B prepares a reply message (m') for A. Just before sending the message to A, B checks its local time (T_3), and includes both times T_2 and T_3 in the reply message
4. A checks the arrival of the reply message from B. As soon as he receives it, A checks its local time (T_4) and writes it down
5. A calculates the deviation between A and B and the precision of the adjustment

Cristian's algorithm



$t(m_t)$: time returned by the server in message m_t

D : time since sending m_{req} and receiving m_t

min : minimum transmission time of a message

Assumption: the server assigns the reference $t(m_t)$ in the middle of the interval $D - 2 * min$ (instant t in the client). This gives the best precision

Deviation: $\theta = t - t(m_t) = t_1 - D/2 - t(m_t)$

Precision = $D/2 - min$

Cristian's algorithm

A similar experiment as the one for NTP can be performed for *Cristian's* algorithm

1. A is the client and B is the server
2. Identify D and $t(m_t)$ in your experiment. Is it possible to determine a value for min ?
3. Calculate the deviation and accuracy according to *Cristian's* algorithm
4. Compare with NTP and justify the results

NTP server hierarchy

```
[sba025@acpt00 sba025]$ /usr/sbin/ntpq acpt00.gi.ehu.es
```

```
ntpq> peers
```

remote	refid	st	t	when	poll	reach	delay	offset	jitter
*scuc06.sc.ehu.es	L0.EB-Bilbao1.r	3	u	510	1024	377	0.918	-0.406	0.235

```
[sba025@acpt00 sba025]$ /usr/sbin/ntpq scuc06.sc.ehu.es
```

```
ntpq> peers
```

remote	refid	st	t	when	poll	reach	delay	offset	jitter
*L0.EB-Bilbao1.r	pulsar.rediris.	2	u	692	1024	377	7.614	-0.456	0.283
-L0.EB-Pamplona1	hora.rediris.es	2	u	227	1024	377	10.401	-0.976	0.122
+158.227.198.253	L0.EB-Bilbao1.r	3	u	292	1024	377	0.909	-0.507	0.681
+L0.EB-Santander	pulsar.rediris.	2	u	564	1024	377	10.477	-0.335	0.009
-scsx01.sc.ehu.e	L0.EB-Bilbao1.r	3	u	33	1024	377	1.460	2.381	0.266
-lgx01.lg.ehu.e	L0.EB-Bilbao1.r	3	u	958	1024	377	5.889	-0.212	0.018

```
[sba025@acpt00 sba025]$ /usr/sbin/ntpq pulsar.rediris.es
```

```
ntpq> peers
```

remote	refid	st	t	when	poll	reach	delay	offset	jitter
*GPS_ONCORE(0)	.GPS.	0	l	1	16	377	0.000	0.001	0.001
+hora.rediris.es	.GPS.	1	u	11	16	376	0.287	0.003	0.006
+hora.roa.es	.GPS.	1	u	10	16	377	15.746	-0.057	0.068
-ntp.roa.es	.GPS.	1	u	6	16	377	51.974	-1.321	1.257

```
ntpq>
```